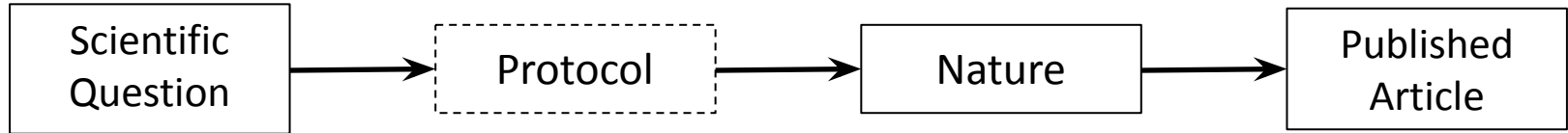


Data Wrangling in R

Reproducible Research

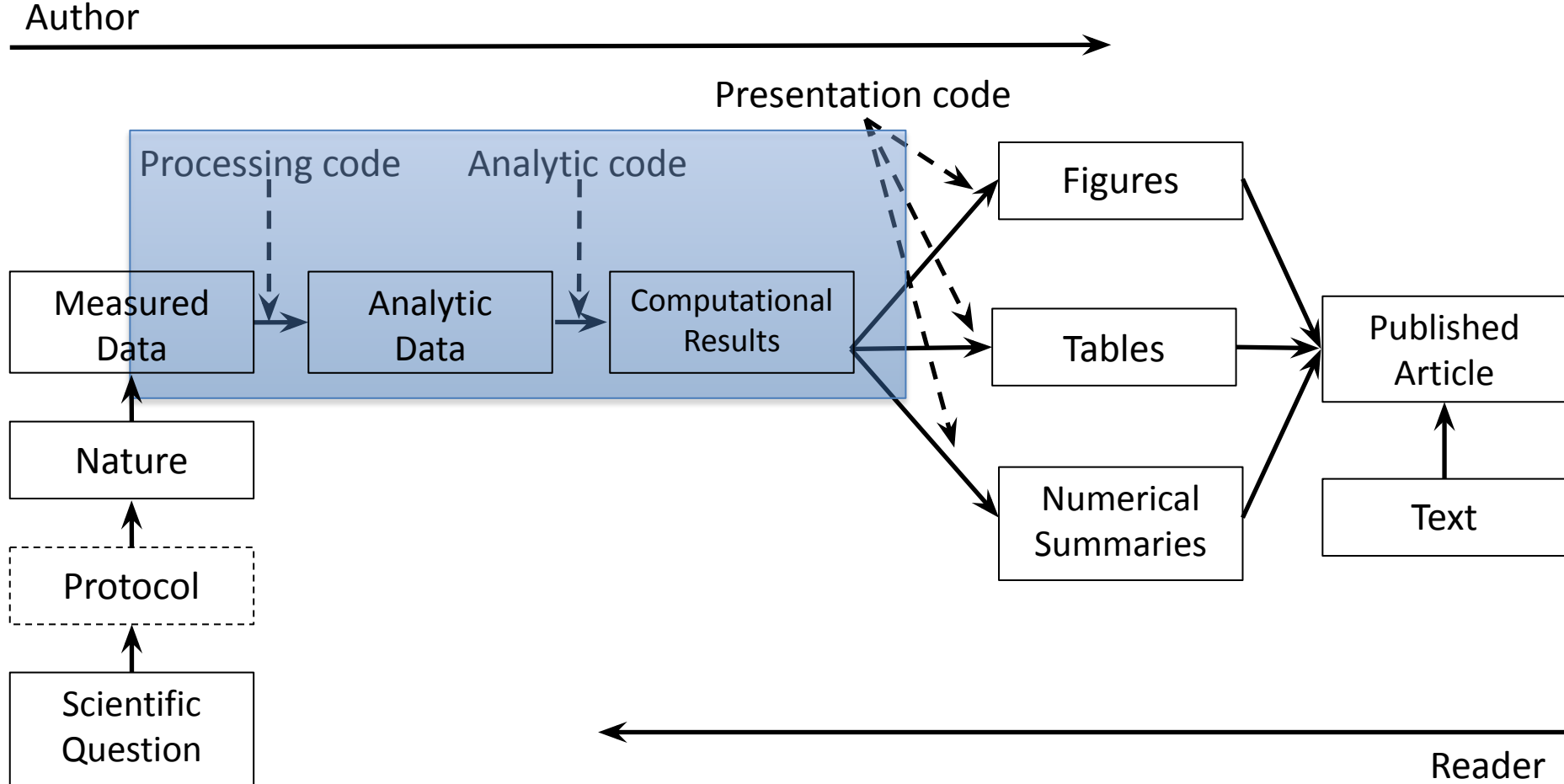
What is Reproducible Research?

Author



Reader

What is Reproducible Research?



Why is this important?

Research Article

Cancer Biology, Computational and Systems Biology

Investigating the replicability of preclinical cancer biology

Timothy M Errington , Maya Mathur, Courtney K Soderberg, Alexandria Denis, Nicole Perfito, Elizabeth Iorns, Brian A Nosek

Center for Open Science, United States; Quantitative Sciences Unit, Stanford University, United States; Science Exchange, United States; University of Virginia, United States

Dec 7, 2021 • <https://doi.org/10.7554/eLife.71601>  



Reproducibility in Cancer Biology: Challenges for assessing replicability in preclinical cancer biology

Timothy M Errington , Alexandria Denis, Nicole Perfito, Elizabeth Iorns, Brian A Nosek

Center for Open Science, United States; Science Exchange, United States;
University of Virginia, United States

Dec 7, 2021 • <https://doi.org/10.7554/eLife.67995>  

“We report the challenges confronted during a large-scale effort to replicate findings in cancer biology, and describe how improving **transparency and sharing** can make it easier to assess rigor and replicability and, therefore, to **increase research efficiency.**”

1. Code + documentation
2. Versions of software
3. Data provenance



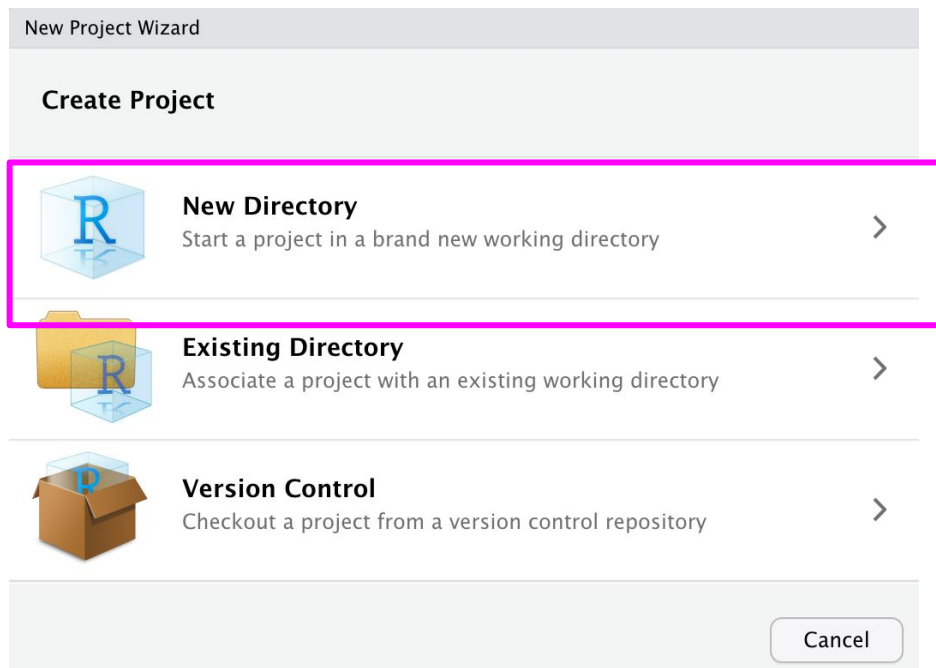
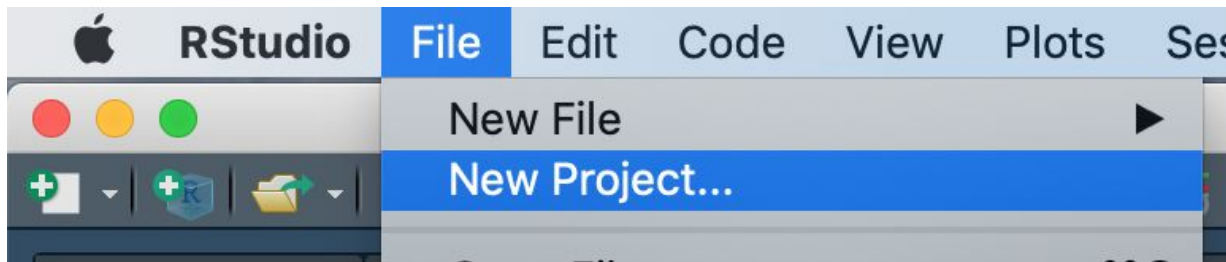
<https://tenor.com/view/struggle-cant-move-over-it-hard-no-gif-4734482>

Your closest collaborator is
you six months ago, but you
don't reply to emails

- Karl Broman

(http://kbroman.org/Tools4RR/assets/lectures/06_org_eda.pdf)

RStudio Projects



New Project Wizard

Back

Project Type



New Project



New Project Wizard

Back

Create New Project



Directory name:

SISBID

Create project as subdirectory of:

~/Desktop

Browse...

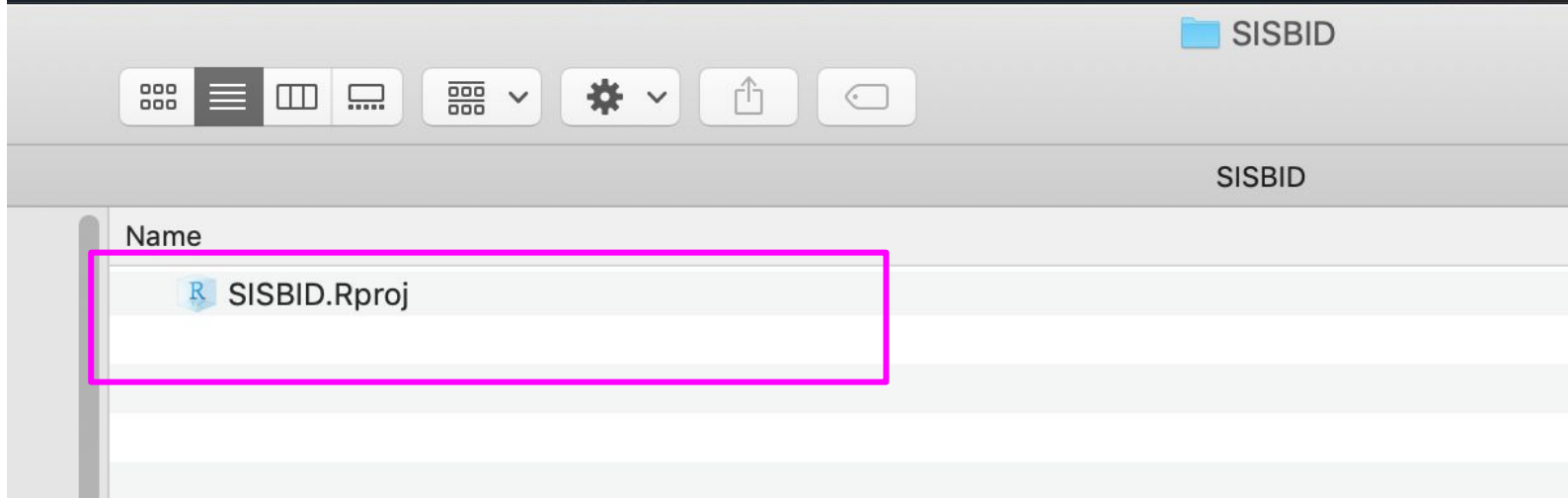
☒ Create a git repository

☐ Use renv with this project

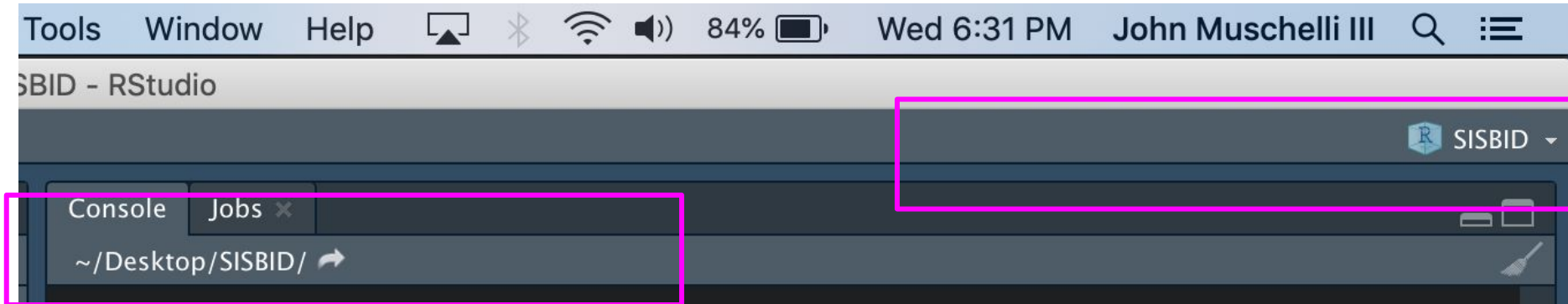
☐ Open in new session

Create Project

Cancel



Double click on the Rproj file - opens RStudio



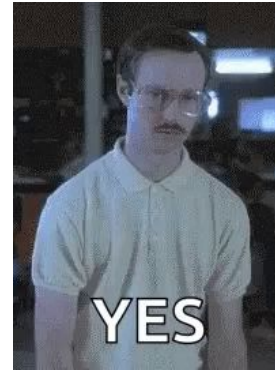
RStudio Projects

- Rproj - Open to directory of Project
- Everything is **relative** to working directory
- Zip up whole folder and send to someone else

```
read_csv("C:/terrible/path/john/blah.csv")
```



```
read_csv("data/blah.csv")
```



R Markdown lets you test your work

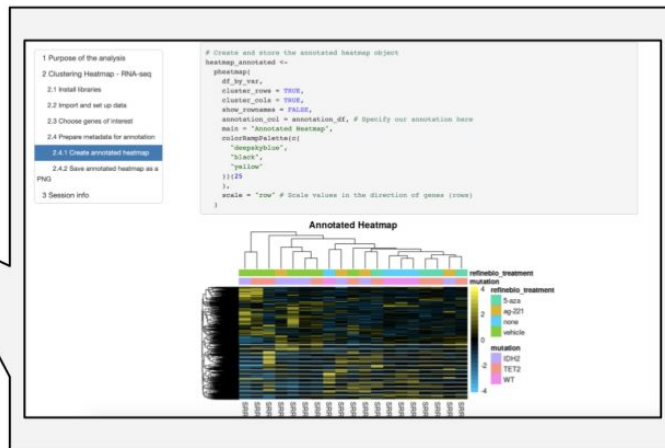
Working from this notebook allows me to interactively develop on my data analysis and write down my thoughts about the process all in one place!



Ruby the
Researcher

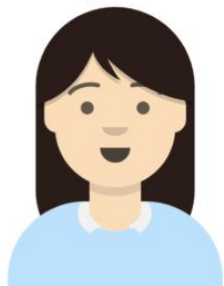


**RMarkdown is conducive to
interactive development!**

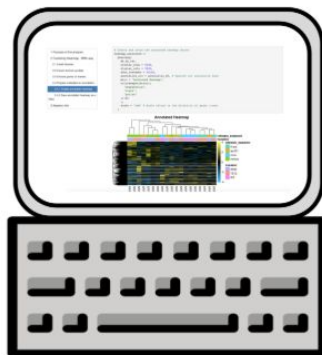


R Markdown allows you to more clearly show what you did

Avi, here's some output from this scientific notebook I've been developing from!



Ruby the
Researcher



This is so easy to follow and read, even though I didn't write the code. Thanks for sharing your exciting results!



Avi the
Associate

**RMarkdown creates easily
shareable output!**



Image created by Candace Savonen using Avataars.

R Markdown makes it easier to update code and see results

Yay! I just got the data for 5 more samples. Because of my handy notebook set up, I can easily call one command and re-run the analysis so it is updated with the new samples included!



Ruby the
Researcher



**RMarkdown is handy for
creating updateable reports!**

The magic of Markdown

- bullets
- **bold**
- *italics*
- [links](https://google.com)
- or run inline `r code`



- bullets
- **bold**
- *italics*
- links
- or run inline `r code`

https://rmarkdown.rstudio.com/authoring_basics.html

Introduction

Here is some background you need to know:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam iaculis enim ut enim viverra molestie. In lacinia aliquet urna, nec vulputate quam congue et. Maecenas porta mauris sem, nec laoreet sapien tincidunt non. Integer sit amet consequat neque, non iaculis ligula.

Hypothesis

Pellentesque molestie erat nec elit efficitur, sit amet sodales erat viverra. Mauris sed commodo eros, ac volutpat sem. Morbi convallis leo et dui cursus, eu suscipit turpis efficitur.

Section 1 code and results

First I will run this.

```
##{r}
print("Hello world")
print("Yup, this is important")
```

The output of which is consistent with my hypothesis.

Conclusion

I can move on to the next part of my project

.Rmd
document

Introduction

Here is some background you need to know:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam iaculis enim ut enim viverra molestie. In lacinia aliquet urna, nec vulputate quam congue et. Maecenas porta mauris sem, nec laoreet sapien tincidunt non. Integer sit amet consequat neque, non iaculis ligula.

Hypothesis

Pellentesque molestie erat nec elit efficitur, sit amet sodales erat viverra. Mauris sed commodo eros, ac volutpat sem. Morbi convallis leo et dui cursus, eu suscipit turpis efficitur.

Section 1 code and results

First I will run this.

```
print("Hello world")
```

```
## [1] "Hello world"
```

```
print("Yup, this is important")
```

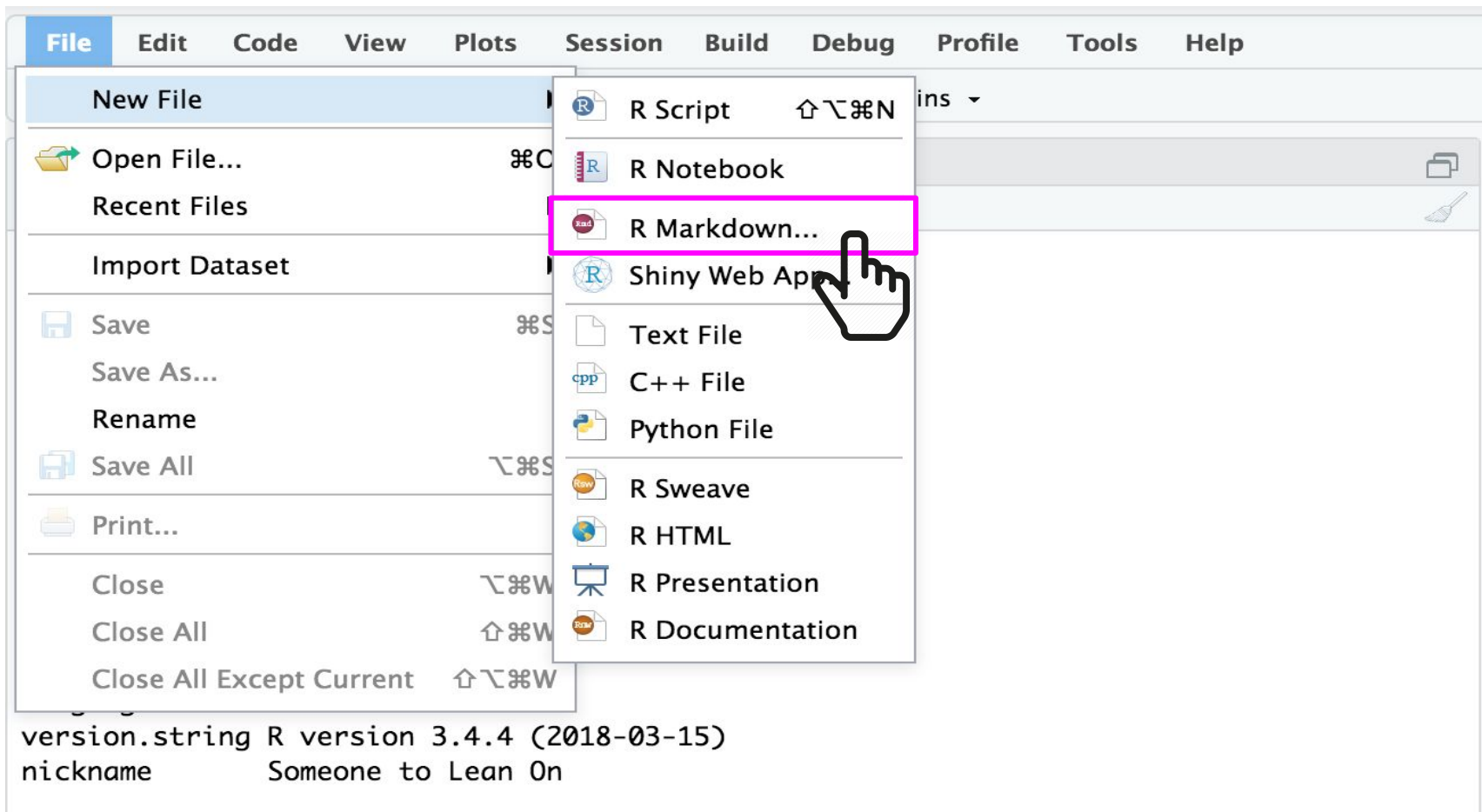
```
## [1] "Yup, this is important"
```

The output of which is consistent with my hypothesis.

Conclusion

I can move on to the next part of my project

PDF, HTML or Word
document



Install Required Packages



Creating R Markdown documents requires updated versions of the following packages: evaluate, highr, markdown, yaml, htmltools, caTools, bitops, knitr, jsonlite, base64enc, rprojroot, rmarkdown.

Do you want to install these packages now?

Yes

No



New R Markdown



Document



Presentation



Shiny



From Template

Title:

Untitled

Author:

Default Output Format:

☒ HTML

Recommended format for authoring (you can switch to PDF or Word output anytime).

☐ PDF

PDF output requires TeX (MiKTeX on Windows, MacTeX 2013+ on OS X, TeX Live 2013+ on Linux).

☐ Word

Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux).

OK

Cancel

New R Markdown

 Document

 Presentation

 Shiny

 From Template

Title:

Author:

Default Output Format:

☒ **HTML**
Recommended format for authoring (you can switch to PDF or Word output anytime).

☐ **PDF**
PDF output requires TeX (MiKTeX on Windows, MacTeX 2013+ on OS X, TeX Live 2013+ on Linux).

☐ **Word**
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux).

YAML - not markdown,
Spaces matter, usually goes
option: value

TEXT

CODE CHUNK

```
1 ---
2 title: "My First R Markdown Document!"
3 author: "Jane Doe"
4 date: "5/31/2018"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word
15 documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.
16
17 When you click the Knit button a document will be generated that includes both content as well as the
18 output of any embedded R code chunks within the document. You can embed an R code chunk like this:
19
20 ```{r cars}
21 summary(cars)
22 ```
23
24 ## Including Plots
25
26 You can also embed plots, for example:
27
28 ```{r pressure, echo=FALSE}
29 plot(pressure)
30 ```
31
32 2:1 # My First R Markdown Document! R Markdown
```


Untitled1 x



```
1 ---
2 title: "My First R Markdown Document!"
3 author: "Jane Doe"
4 date: "5/31/2018"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
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25
26 You can also embed plots, for example:
27
28 ```{r pressure, echo=FALSE}
29 plot(pressure)
30 ```
```

2:1 # My First R Markdown Document! R Markdown

Save File – Untitled1

File name:

> / > cloud > project



..



.Rhistory

0 B

May 30, 2018, 12:34 PM



project.Rproj

205 B

May 31, 2018, 3:35 PM

New Folder

Save

Cancel



Header rendered as the title

My First R Markdown Document!

Jane Doe

5/31/2018

R Markdown

Text section rendered as formatted text

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

Code rendered as the input code *AND* the output of running the code chunk

```
##      speed      dist
##  Min.   : 4.0    Min.   : 2.00
##  1st Qu.:12.0    1st Qu.: 26.00
##  Median :15.0    Median : 36.00
##  Mean   :15.4    Mean   : 42.98
##  3rd Qu.:19.0    3rd Qu.: 56.00
##  Max.   :25.0    Max.   :120.00
```

Including Plots

You can also embed plots, for example:

test_document.Rmd x

Knit

Insert

Run

```
1 ---
2 title: "My First R Markdown Document!"
3 author: "Jane Doe"
4 date: "5/31/2018"
5 output: html_document
6 ---
7
8 ```{r setup, include=FALSE}
9 knitr::opts_chunk$set(echo = TRUE)
10 ```
11
12 ## R Markdown
13
14 This is an R Markdown document. Markdown is a simple formatting syntax for
15 authoring HTML, PDF, and MS Word documents. For more details on using R Markdown
16 see <http://rmarkdown.rstudio.com>.
17:1 R Markdown
```

Console Terminal x

/cloud/project/

> |

Environment History Connections

Files Plots Packages Help Viewer

New Folder Upload Delete Rename More

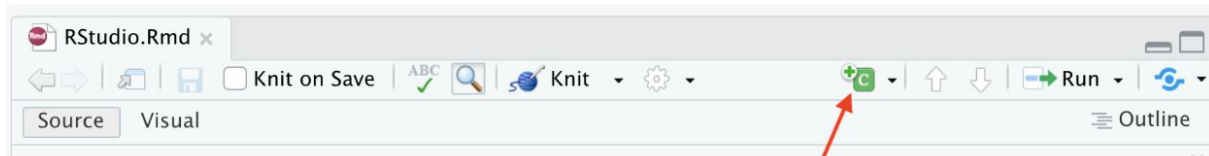
/ > cloud > project

	Name	Size	Modified
	..		
☐	.Rhistory	0 B	May 30, 2018, 12:34 PM
☐	project.Rproj	205 B	Jun 1, 2018, 8:33 AM
☐	test_document.html	741.2 KB	May 31, 2018, 3:57 PM
☐	test_document.Rmd	854 B	May 31, 2018, 3:57 PM

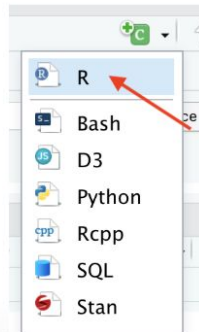
Create Chunks

To create a new R code chunk:

- Use the insert code chunk button at the top of RStudio.

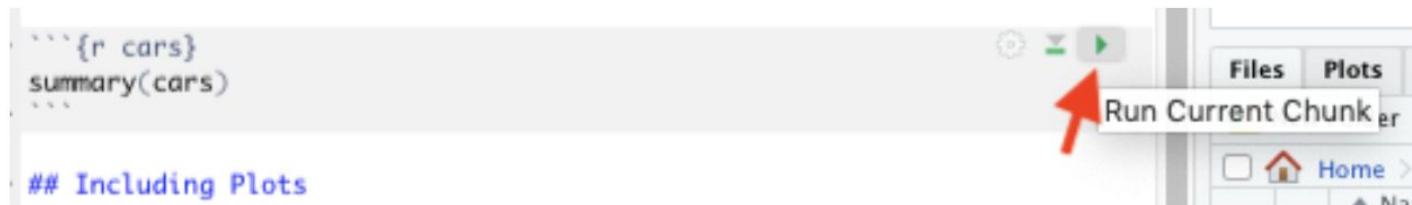


- Select R (default) as the language:



Run code in a chunk

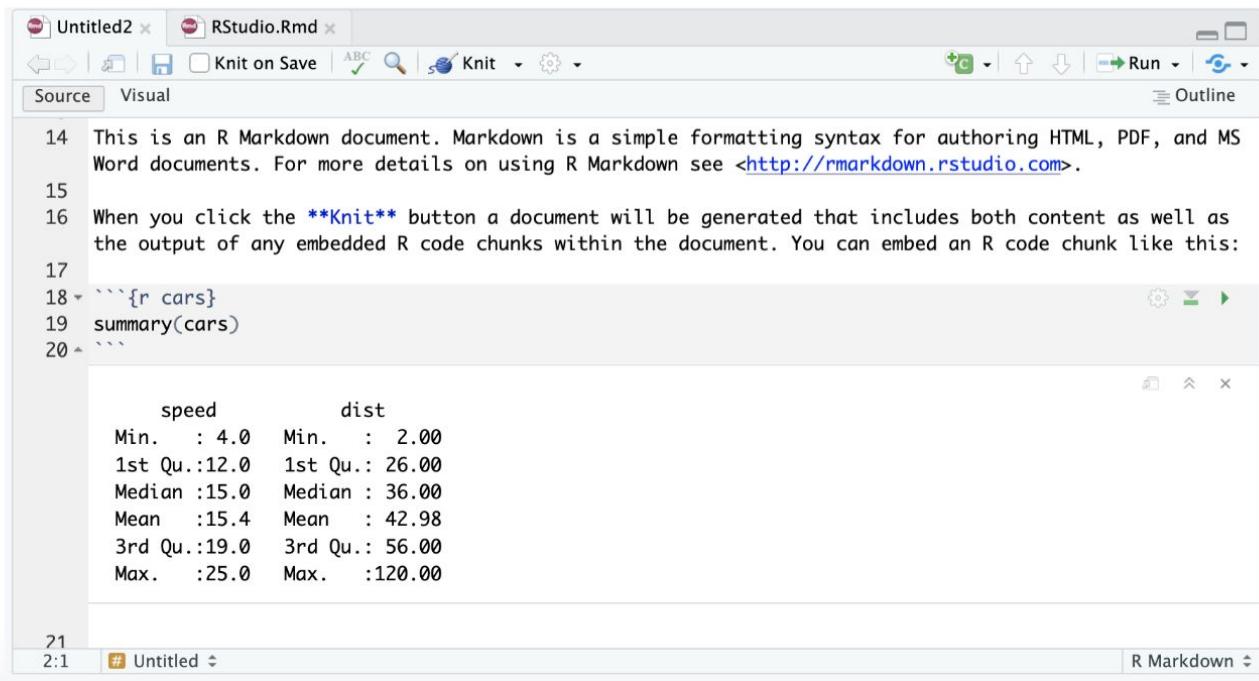
Clicking the run (play) button runs the code in the chunk.



Ctrl + Enter on Windows or Command + Enter on Mac in your script evaluates that line of code

Running a chunk executes the code

- generally see a preview of the output of the code just below the chunk
- see the code in the console



The screenshot shows the RStudio interface with a file named 'RStudio.Rmd' open. The 'Source' pane displays the R Markdown document content. The document includes a title 'Untitled2', a subtitle 'RStudio.Rmd', and a code chunk. The code chunk contains the following R code:

```
summary(cars)
```

The output of the code chunk is displayed in a separate pane below the code, showing the summary of the 'cars' dataset. The output is a table with two columns: 'speed' and 'dist'. The rows represent the minimum, first quartile, median, mean, third quartile, and maximum values for each variable.

speed	dist
Min. : 4.0	Min. : 2.00
1st Qu.: 12.0	1st Qu.: 26.00
Median : 15.0	Median : 36.00
Mean : 15.4	Mean : 42.98
3rd Qu.: 19.0	3rd Qu.: 56.00
Max. : 25.0	Max. : 120.00

The status bar at the bottom of the RStudio window shows the file name 'Untitled2' and the file type 'R Markdown'.

Run previous chunks button

You can run all chunks above a specific chunk using this button:

```
```{r, out.width = "80%", echo = FALSE, fig.align='center'}  
knitr::include_graphics("images/chunk.png")
```
```

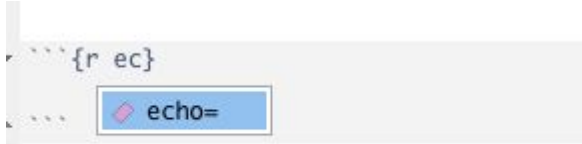



```
{r eval}
```



eval = FALSE will show but not run the code

```
{r ec}
```



echo = FALSE will not show but run the code

Using both set to FALSE will hide the code and will not run the code!

The default is TRUE - do not need to write it out if you want code to be shown and run.

Biiconductor_intro.Rmd x Data_Subsetting_Part1.R x Data_Cleaning.Rmd x

Knit on Save ABC Knit Run

Source Visual Outline

```
60
61 ```{r}
62 p = problems(ufo)
63 p
64 ```
65
66 ## Let's just drop those problematic rows for now.
67
68 Though you would usually want to check them!
69
70 ```{r}
71 ufo = ufo[-p$row,] # brackets can also be used for subsetting
72 ```
```

Environment History Connection

R Global Environment 99 MiB

Chunk Name: Unnamed chunk

Output: (Use document default)

☐ Show warnings

☐ Show messages

☐ Use paged tables

☐ Use custom figure size

[? Chunk options](#)

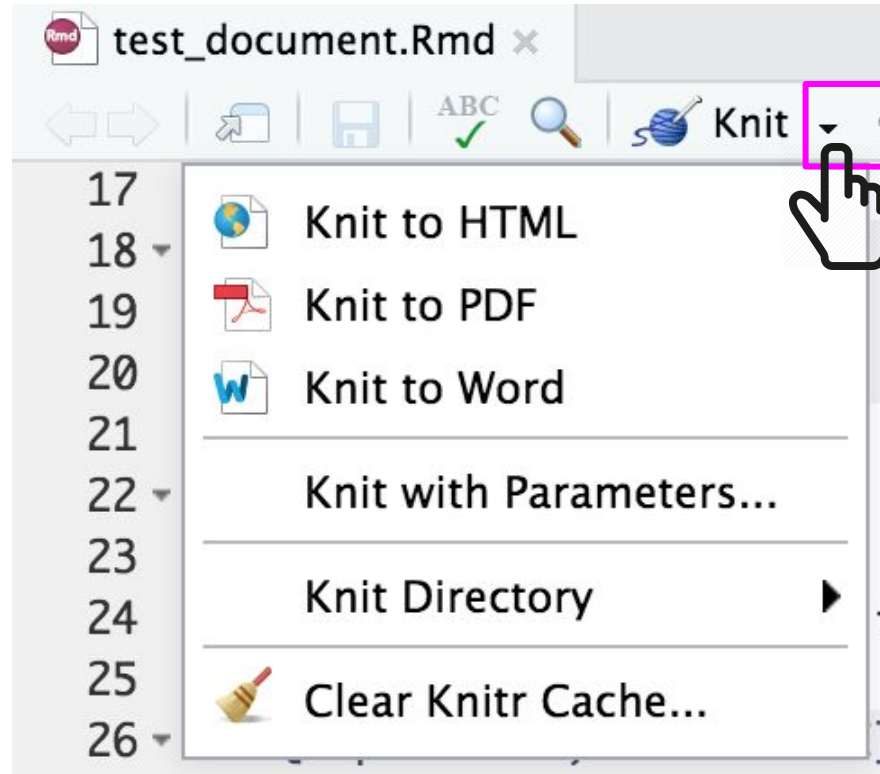
Revert Apply

Can Also influence how a plot is displayed with a chunk!

```
fig.align = "center",  
fig.height = 3, fig.width = 6
```

```
```{r pressure, echo=FALSE, fig.width=6, fig.height=3}  
plot(pressure)
```
```





Gut Check

Where does code go typically in an Rmd file?

A

```
```{r}
```



**B**

```
```
```

C

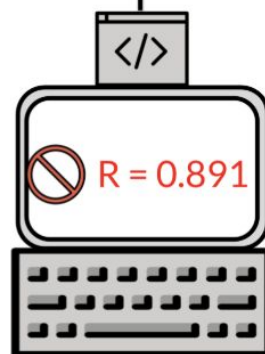
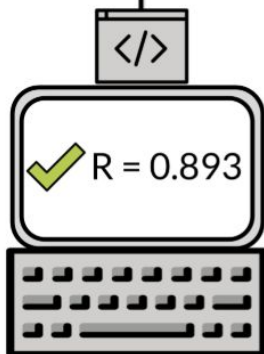
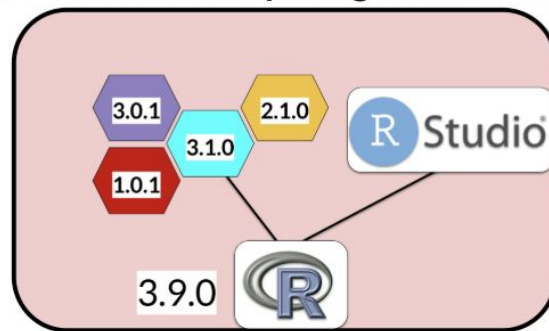
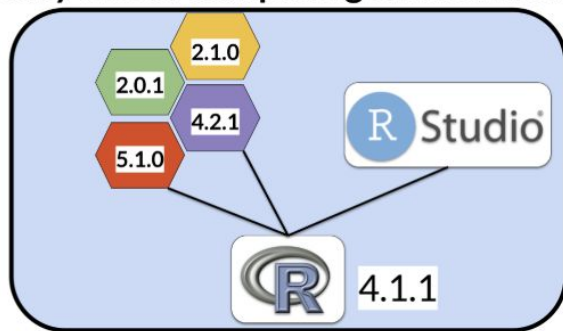
<https://support.rstudio.com/hc/en-us/articles/200711853-Keybaord-Shortcuts>

Rendering in R

```
library(rmarkdown)  
render("Untitled.Rmd")
```

Versions matter

Ruby's local computing environment Avi's local computing environment



Session information - what's loaded?

```
install.packages("devtools")  
library(devtools)  
session_info()
```

Rmarkdown lab

Download the file from

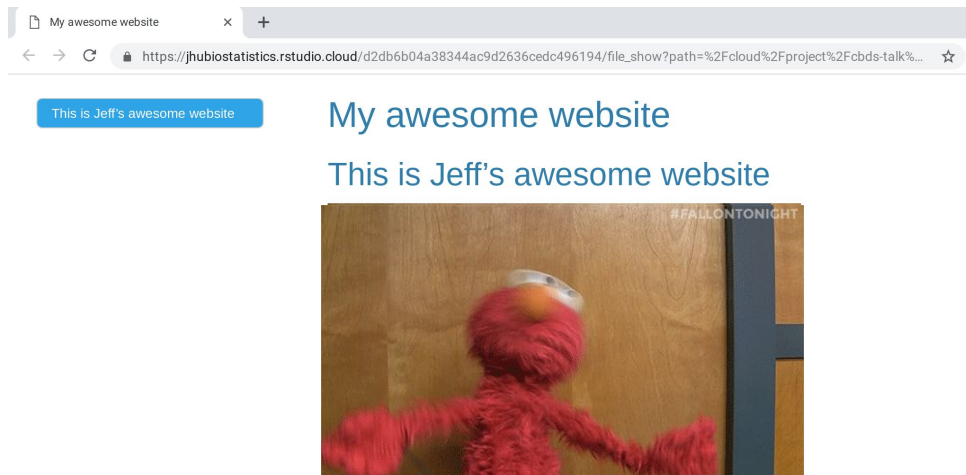
<https://github.com/SISBID/data-wrangling/raw/gh-pages/labs/rmarkdown-lab.Rmd>

Barely scratching the surface

rmarkdown

```
---  
title: "My awesome website"  
output:  
  html_document:  
    toc: true  
    toc_float: true  
    theme: cerulean  
---  
# This is Jeff's awesome website  
  

```



flexdashboard

title: "How does your BMI measure up?"

output: flexdashboard::flex_dashboard

runtime: shiny

Inputs {.sidebar}

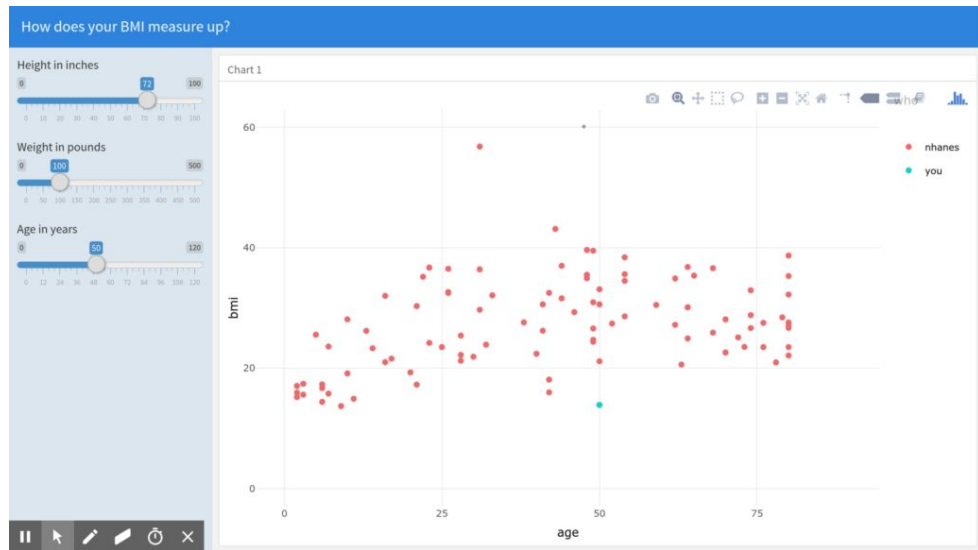
```
```{r}
library(flexdashboard); library(NHANES); library(plotly);library(dplyr)
sliderInput("height", "Height in inches",0,100,72)
sliderInput("weight", "Weight in pounds",0,500,100)
sliderInput("age", "Age in years",0,120,50)
```

...

Column

### Chart 1

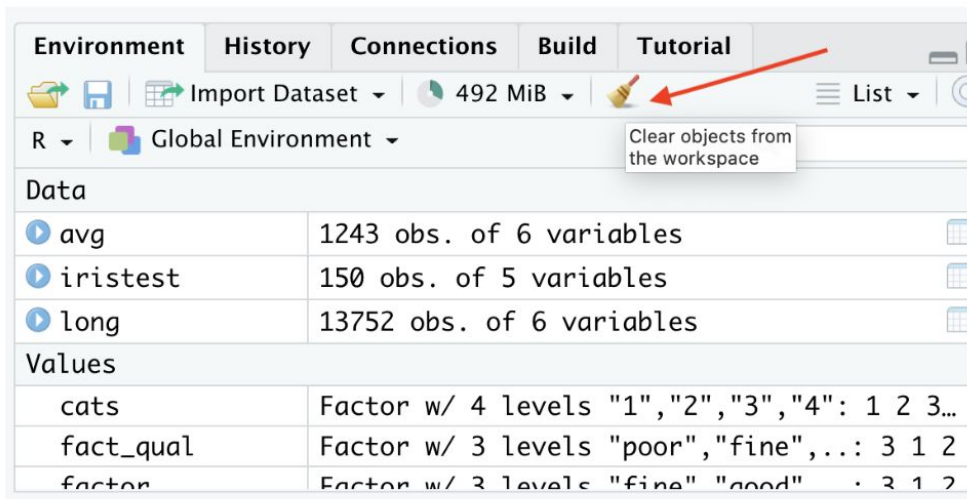
```
```{r}
nhanes = sample_n(NHANES,100)
renderPlotly({
  df = data.frame(bmi = c(nhanes$BMI,input$weight*0.45/(input$height*0.025)^2),
    age = c(nhanes$Age,input$age),
    who = c(rep("nhanes",100),"you"))
  ggplotly(ggplot(df) +
    geom_point(aes(x=age,y=bmi,color=who)) +
    scale_x_continuous(limits=c(0,90)) +
    scale_y_continuous(limits=c(0,60)) +
    theme_minimal()
  )
})
```
```



# Clean your environment

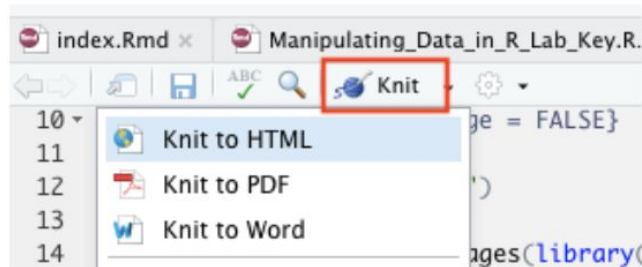
Regularly cleaning your environment and trying your code again, can help ensure that your code is running as expected.

Occasionally we might forget to save a step of our code in our R Markdown file that we ran only in the console. This will help us figure that out.



# Check if your file knits regularly

Regularly checking if your file knits will help you spot a missing step or error earlier when you have less code to try to identify where your code might have gone wrong.



# GUT CHECK

Why is reproducibility so important?

- A. It helps to ensure that your code is working consistently and it helps others understand what you did
- B. It ensures that your code is correct



# GUT CHECK

What is NOT a practice to improve the reproducibility of our work?

- A. Using R Markdown files to describe what your code is doing
- B. Using scripts instead of R Markdown files
- C. Testing your code with R Markdown files or the run previous button
- D. Regularly cleaning the environment

# Summary

To help make your work more reproducible:

- Use R Projects
- RMarkdown
- Clean your environment regularly
- Check the knit of your RMarkdown regularly
- Tell your future self and others what you did!
- Print session info!

## More resources

These are just some quick tips, for more information:

- [Reproducibility in Cancer Informatics course](#)
- [Advanced Reproducibility in Cancer Informatics course](#)
- [The RMarkdown book](#)
- [Jenny Bryan's organizational strategies.](#)
- [Write efficient R code for science.](#)